

Matthew Dolin

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EDUCATION

Master of Science in Data Science

Drexel University
GPA: 4.0

Jun 2027 (Expected)
Philadelphia, PA

Bachelor of Science in Computer Science

Lehigh University
Minors: Data Science, Cognitive Science

May 2024
Bethlehem, PA
GPA: 3.45

SKILLS

- **Programming Languages:** Python | SQL | R | Java | JavaScript
- **Data Analysis & Machine Learning:** Pandas | NumPy | Scikit-Learn | Statistical Analysis | Data Cleaning
- **Data Visualization & BI:** Excel | Plotly | Power BI | Tableau | Dashboard Development | Data Storytelling
- **Version Control:** Git | GitHub | Merge Conflict Resolution | Code Review Workflows
- **Certifications:** Career Essentials in Data Science - Microsoft and LinkedIn (May 2025)

DATA SCIENCE PROJECTS

Recipe Recommender Web App

Lead Django Developer

Drexel University
Jun 2026

- Trained and evaluated recommender models using Python and Lenskit to generate personalized recipe recommendations
- Implemented dynamic UI updates with jQuery and AJAX, allowing users to filter by dietary restrictions without full page reloads
- Engineered a robust Django backend to manage user profiles, recipe attributes, and algorithmic recommendation logic
- Designed a relational database schema using SQLite, leveraging indexes and foreign keys to optimize query execution times

Natural Gas Trading Optimization (Forage/JP Morgan Chase)

Independent Data Science Project

GitHub
Aug 2025

- Analyzed seasonal and historical price trends in natural gas markets to identify profit-maximizing purchase and sale strategies
- Built a forecasting pipeline using cubic spline interpolation and ARIMA time-series modeling to generate daily natural gas price predictions 12 months ahead
- Implemented constrained optimization logic for storage trading strategies under injection, withdrawal, and capacity constraints

Equitable Energy Capstone

Data Engineering Lead

Lehigh University
Jan 2023 - Dec 2023

- Designed and implemented an ETL pipeline to collect and structure Energy Justice Community data for analysis of utility pricing disparities
- Developed Python web crawlers using BeautifulSoup and Requests to aggregate historical Pennsylvania electricity rate data into a relational SQL database
- Mentored teammates on machine learning best practices and contributed to technical documentation

PROFESSIONAL EXPERIENCE

Lehigh University

Head Teaching Assistant

Bethlehem, PA
Aug 2023 - Dec 2023

- Guided students on completing CamJam EduKit robotics lab assignments through weekly Python code demonstrations
- Coordinated grading responsibilities among my section's teaching assistants using Google Sheets to optimize feedback response times on students' assignments
- Analyzed student performance trends to identify at-risk students and provide targeted technical mentoring

Lehigh University

Teaching Assistant

Bethlehem, PA
Aug 2021 - May 2023

- Evaluated students' assignments using Lehigh's learning management system and Jupyter Notebook to assess their understanding of programming concepts
- Mentored students on Python best practices and useful documentation resources to improve academic outcomes
- Aided students in debugging code and clarifying assignment requirements during weekly office hours